Managerial Economics (ARE) 106
University of California, Davis
Instructor: John H. Constantine
Discussion Section 1

Problem 1: Use the accompanying pages to answer the Problem 1 questions.

(a) Use the attached Excel sheet to calculate b_1 and b_2. Numbers are expected.
(b) Write the OLS equation given your answers in part (a).
(c) Verify that the OLS equation passes through the means of the data ($\bar{x}, \bar{y}$).
(d) Calculate $\hat{y}_i$ for each of the x_i.
(e) Calculate $\hat{e}_i$ for each of the x_i. What is the value of $\sum_1^5 \hat{e}_i^2$?
(f) On the graph, show that the mean values indeed fall on the regression line?
(g) On the graph, draw in the $\hat{e}_i$ for each of the x_i.
(h) Show that the errors are homoscedastic.
(i) Calculate $\hat{\sigma}^2$. $\hat{\sigma}^2$

Problem 2:
State whether the following models are linear regression models.

(a) $y_i = \beta_1 + \beta_2 x_i + e_i$.
(b) $y_i = \beta_1 + \beta_2(1/x_i) + e_i$.
(c) $y_i = \beta_1 + \beta_2 \ln(x_i) + e_i$.
(d) $\ln(y_i) = \beta_1 + \beta_2 x_i + e_i$.
(e) $\ln(y_i) = \beta_1 + \beta_2 \ln(x_i) + e_i$.
(f) $y_i = \beta_1 + \beta_2\beta_3 x_i + e_i$.
(g) $y_i = \beta_1 + \beta_2^2 x_i + e_i$.

Problem 3:
State whether the following are true, false, or uncertain.

(a) The stochastic error term $\hat{e}_i$ and the residual term e_i mean the same thing.
(b) In OLS, the regression the regression coefficients, the βs are random variables and bs are parameters.

(a) Calculate b_1 & b_2.

0 −2

Problem 1:

	B	C	D	E	F	G
1	x	y	x – xbar	$(x - xbar)^2$	y – ybar	(x - xbar)(y – ybar)
2	0	6	-2	4	3.6	-7.2
3	1	2	-1	1	-0.4	0.4
4	2	3	0	0	0.6	0
5	3	1	1	1	-1.4	-1.4
6	4	0	2	4	-2.4	-4.8
7						
8	**∑xi**	**∑yi**	**∑(x – xbar)**	**∑(x – xbar)2)**	**∑(y – ybar)**	**∑(x - xbar)(y – ybar)**
9						
10						
11	10	12	0	10	0	−13
12						
13						
14						
15	xbar	ybar				
16	10/5=2	12/5=2.4				
17						
18						
19	x	y	yhat	ehat	$ehat^2$	xehat
20	0	6	5	1	1	0
21	1	2	3.7	-1.7	2.89	-1.7
22	2	3	2.4	0.6	0.36	1.2
23	3	1	1.1	-0.1	0.01	-0.3
24	4	0	-0.2	0.2	0.04	0.8
25						
26	**∑xi**	**∑yi**	**∑yhat**	**∑ehat**	**$\sum ehat^2$**	**∑xehat**
27						
28						
29						
30						
31						
32						
33	b_1	b_2				
34						

−7.2
−5.8

$$\text{Slope} = b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{-13}{10} = -1.3$$

$$y_i = b_1 + b_2 x_i + \hat{e}_i$$

$$\bar{y} = b_1 + b_2 \bar{x} \Rightarrow b_1 = \bar{y} - b_2 \bar{x}$$

$$\Rightarrow b_1 = 2.4 - (-1.3 * 2) = 2.4 + 2.6 = 5$$

$$\boxed{\hat{Y}_i = 5 - 1.3 X_i}$$

$\hat{Y}_i = 5 - 1.3X_i$

Problem 1:

	B	C	D	E	F	G
1	x	y	x – xbar	$(x - xbar)^2$	y – ybar	(x - xbar)(y – ybar)
2	0	6	-2	4	3.6	-7.2
3	1	2	-1	1	-0.4	0.4
4	2	3	0	0	0.6	0
5	3	1	1	1	-1.4	-1.4
6	4	0	2	4	-2.4	-4.8
7	∑xi	∑yi	∑(x – xbar)	∑(x – xbar)2)	∑(y – ybar)	∑(x - xbar)(y – ybar)
8						
9						
10	10	12	0	10	0	-13
11						
12						
13						
14						
15	xbar	ybar				
16	10/5=2	12/5=2.4				
17						
18						
19	x	y	yhat	ehat	$ehat^2$	xehat
20	0	6	5	1	1	0
21	1	2	3.7	-1.7	2.89	-1.7
22	2	3	2.4	0.6	0.36	1.2
23	3	1	1.1	-0.1	0.01	-0.3
24	4	0	-0.2	0.2	0.04	0.8
25	∑xi	∑yi	∑yhat	∑ehat	$\sum ehat^2$	∑xehat
26						
27						
28	10	12	12	0	4.3	0
29						
30						
31						
32						
33	b_1	b_2				
34	5	-1.3				

-7.2
-5.8

$Y_i - \hat{Y}_i$

residual

ehat

$Y_i = 5 - 1.3X_i + \hat{e}_i$

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.892843666
R Square	0.797169811
Adjusted R Square	0.729559748
Standard Error	1.197219
Observations	5

ANOVA

	df	SS	MS	F	Significance F
Regression	1	16.9	16.9	11.79069767	0.041424185
Residual	3	4.3	1.433333333		
Total	4	21.2			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	5	0.92736185	5.39163866	0.012501997	2.048720708	7.951279292
x	-1.3	0.37859389	-3.433758535	0.041424185	-2.504854726	-0.095145274

b_1 b_2

RESIDUAL OUTPUT

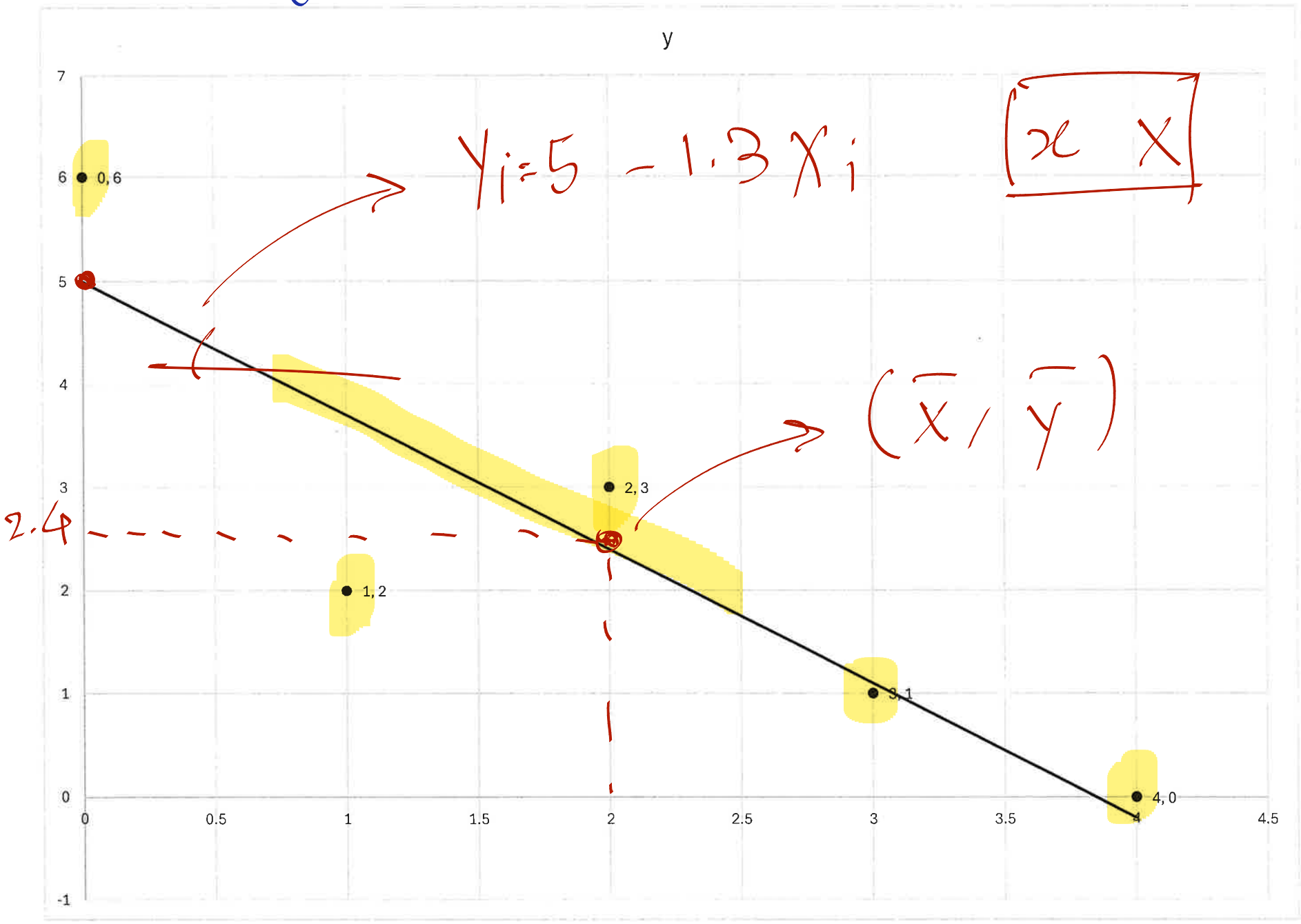
Observation	Predicted y	Residuals
1	5	1
2	3.7	-1.7
3	2.4	0.6
4	1.1	-0.1
5	-0.2	0.2

(b) write down the OLS equation.

$$Y_i = 5 - 1.3X_i + \hat{e}_i$$

$$\hat{Y}_i = 5 - 1.3X_i$$

(c) verify as passes through mean $\bar{x}, \bar{y}$



(a) calculate $\hat{y}_i$ for each x_i

use $\hat{y}_i = 5 - 1.3\, x_i$ for $x_i = \begin{cases} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{cases}$

please look at the table!

(e) Calculate $\hat{e}_i$ for each x_i. What is $\sum_i \hat{e}_i^2$?

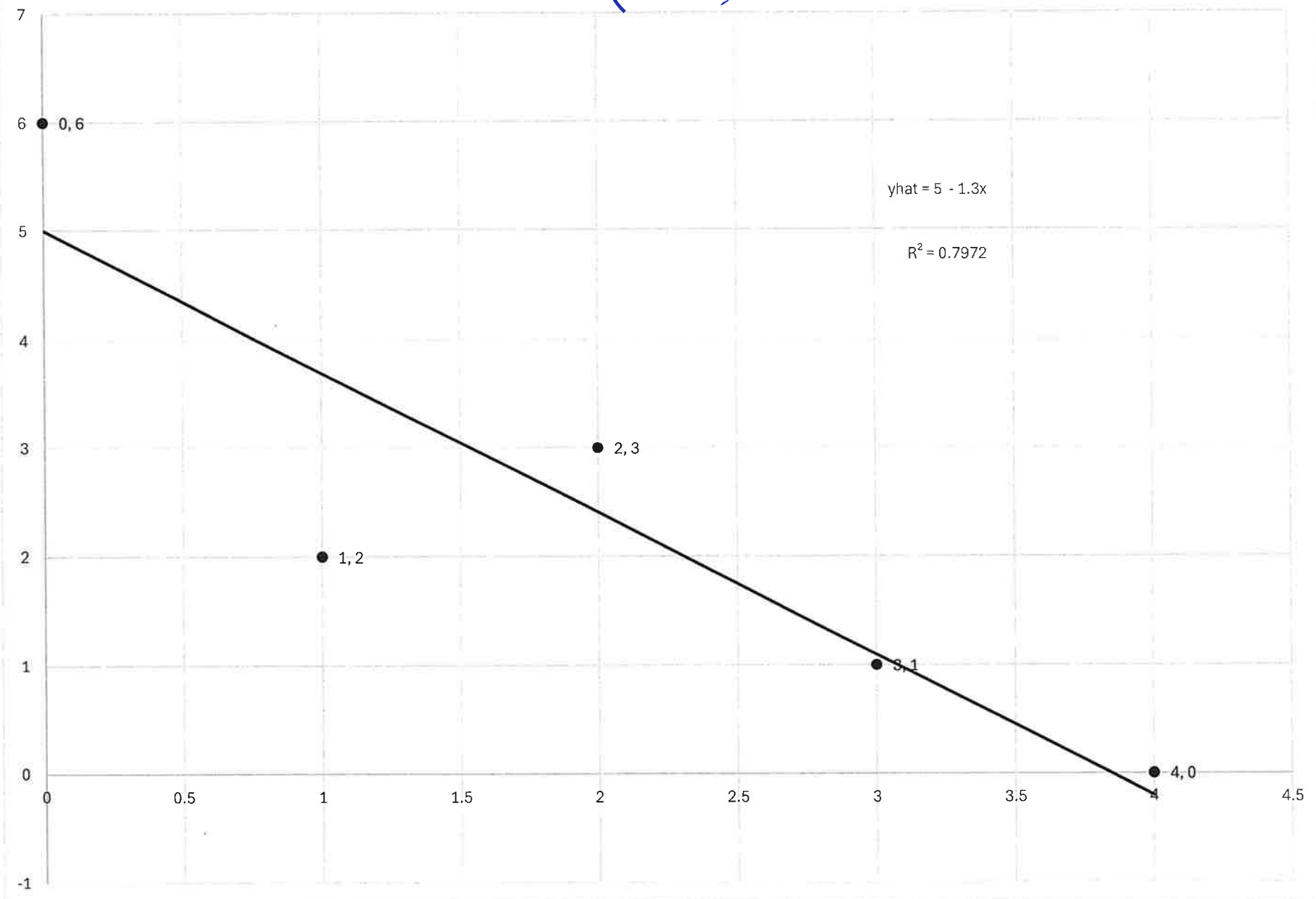
residual $:= y_i - \hat{y}_i$

find $y_i - \hat{y}_i$ for $y_i = \begin{cases} 6 \\ 2 \\ 3 \\ 1 \\ 0 \end{cases}$

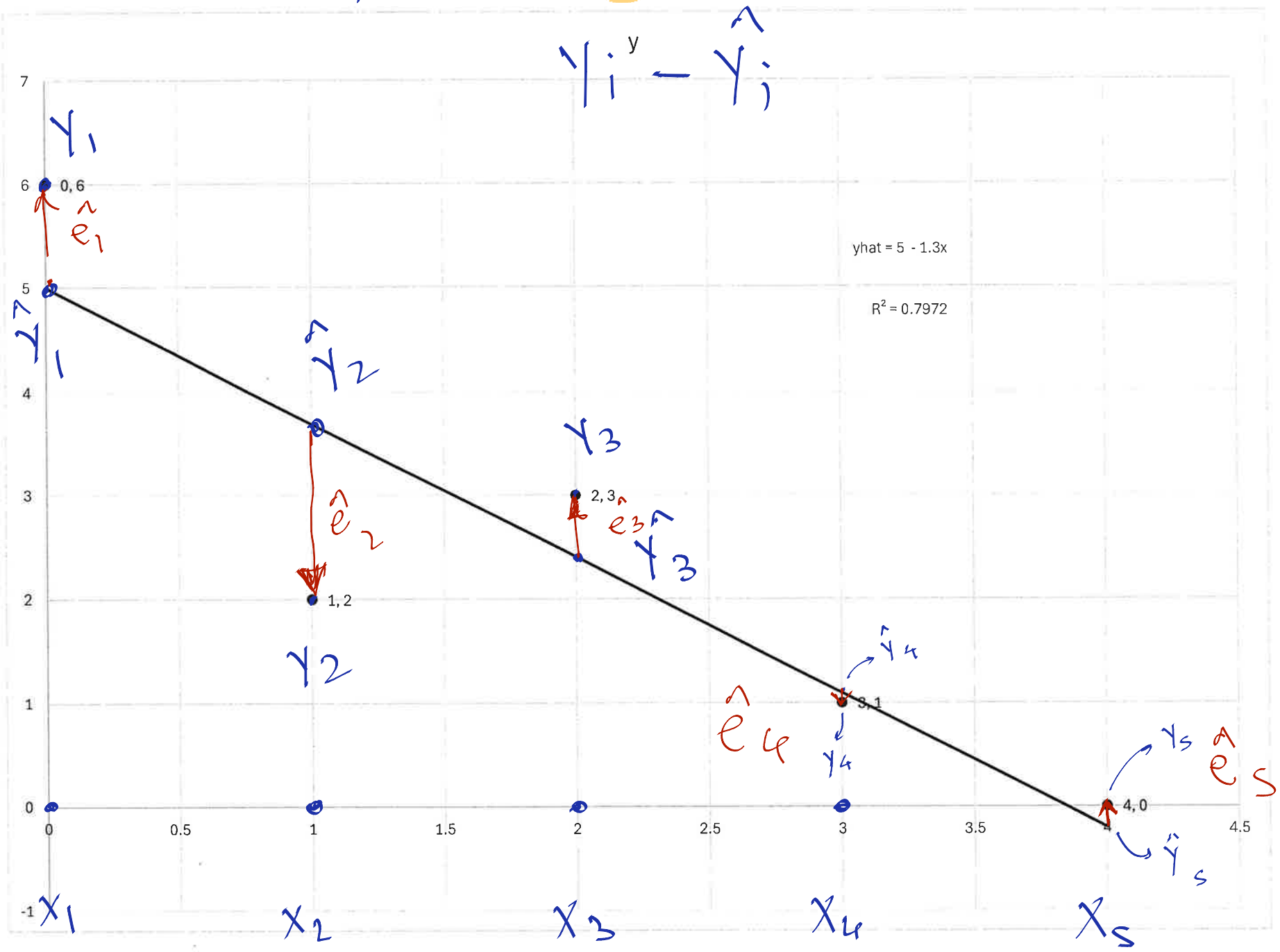
Please look at the table.

$\sum \hat{e}_i^2 = 4.3$ (based on table)

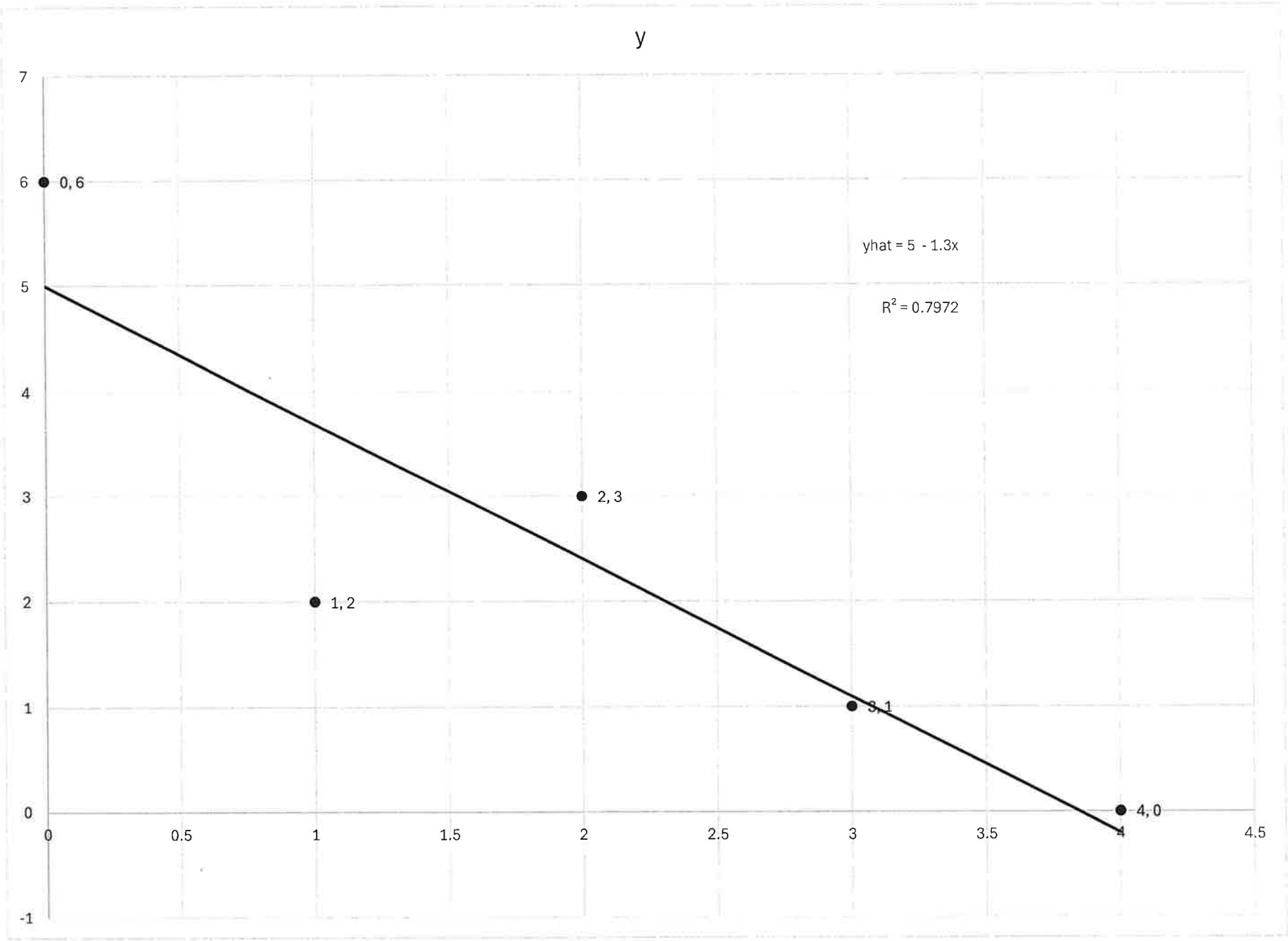
(f) on the graph show mean values fall on regression line → same as (c)

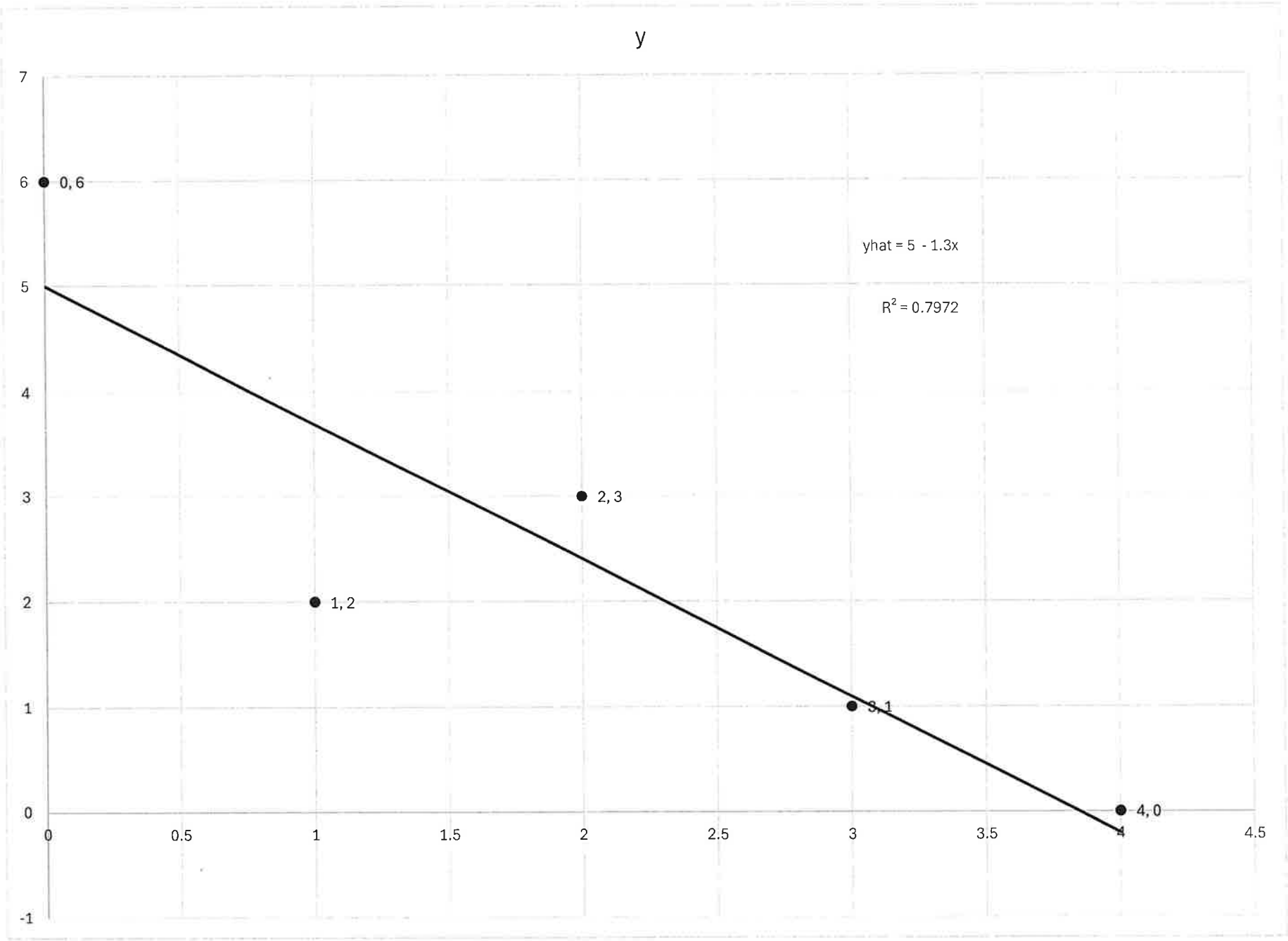


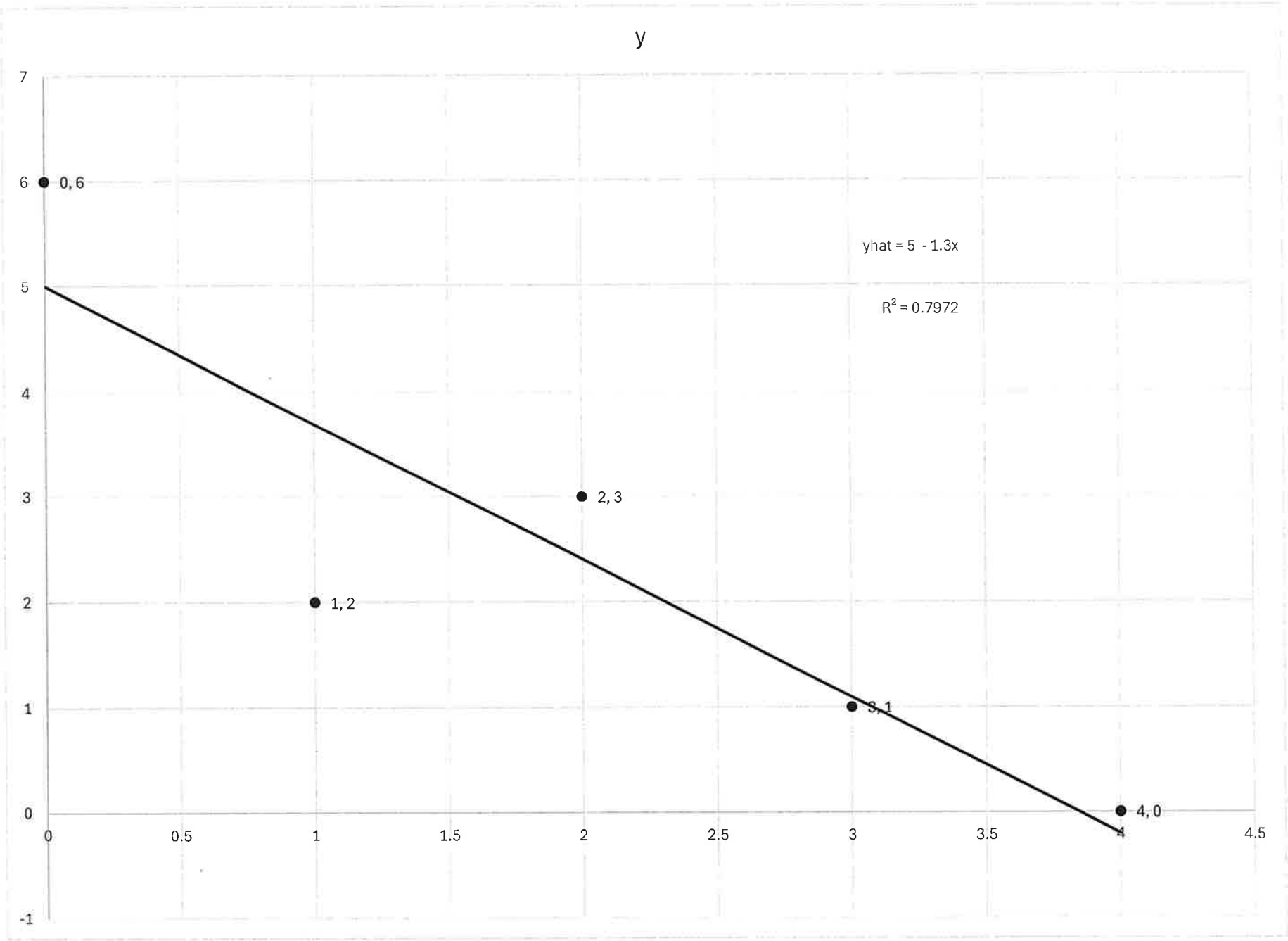
(4) on the graph draw $\hat{e}_i$ for each x_i.



(h) $\longrightarrow$ skip


y
7
6
5
4
3
2
1
0
-1
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
0, 6
2, 3
1, 2
3, 1
4, 0
yhat = 5 - 1.3x
R² = 0.7972


y
7
6
5
4
3
2
1
0
-1
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
0, 6
2, 3
1, 2
3, 1
4, 0
yhat = 5 - 1.3x
R² = 0.7972


y
7
6
5
4
3
2
1
0
-1
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
0, 6
2, 3
1, 2
3, 1
4, 0
yhat = 5 - 1.3x
R² = 0.7972

(i) Calculate $\hat{\sigma}^2$

2.7 Estimating the Variance of the Error Term

The variance of the random error term, σ^2, is the one unknown parameter of the simple linear regression model that remains to be estimated.

The conditional variance of the random error e_i is

$$\operatorname{var}(e_i|\mathbf{x}) = \sigma^2 = E\left\{\left[e_i - E(e_i|\mathbf{x})\right]^2 \middle| \mathbf{x}\right\} = E(e_i^2|\mathbf{x})$$

if the assumption $E(e_i|\mathbf{x}) = 0$ is correct. Since the "expectation" is an average value, we might consider estimating σ^2 as the average of the squared errors

$$\hat{\sigma}^2 = \frac{\sum e_i^2}{N}$$

This formula is unfortunately of no use since the random errors e_i are *unobservable*! However, although the random errors themselves are unknown, we do have an analog to them—namely, the least squares residuals. Recall that the random errors are

$$e_i = y_i - \beta_1 - \beta_2 x_i$$

From (2.6) the least squares residuals are obtained by replacing the unknown parameters by their least squares estimates:

$$\hat{e}_i = y_i - \hat{y}_i = y_i - b_1 - b_2 x_i$$

It seems reasonable to replace the random errors e_i by their analogs, the least squares residuals, so that

$$\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N}$$

This estimator, though quite satisfactory in large samples, is a *biased* estimator of σ^2. But there is a simple modification that produces an unbiased estimator:

$$\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} \tag{2.19}$$

$$= 4.3 \cancel{(5-2)} = \frac{4.3}{3}$$

$$= 1.43$$

The 2 that is subtracted in the denominator is the number of *regression parameters* (β_1, β_2) in the model, and this subtraction makes the estimator $\hat{\sigma}^2$ unbiased, so that $E\left(\hat{\sigma}^2|\mathbf{x}\right) = \sigma^2$.

Problem 3:

State whether the following are true, false, or uncertain.

(a) The stochastic error term $\hat{e}_i$ and the residual term e_i mean the same thing.

(b) In OLS, the regression the regression coefficients, the βs are random variables and bs are parameters.

residual

stochastic error term from econometric model

FALSE

FALSE

Econometric model $y_i = \underline{\underline{\beta_1}} + \underline{\underline{\beta_2}} X_i + e_i$

parameter parameter

we estimate parameters using OLS and we call them b_1 b_2

In reality, many other factors may affect household expenditure on food; the ages and sexes of the household members, their physical size, whether they do physical labor or have desk jobs, whether there is a party following the big game, whether it is an urban or rural household, whether household members are vegetarians or into a paleo-diet, as well as other taste and preference factors ("I really like truffles") and impulse shopping ("Wow those peaches look good!"). Lots of factors. Let $e = \textit{everything else}$ affecting food expenditure other than income. Furthermore, even if a household has a rule, strict or otherwise, we do not know it. To account for these realities, we suppose that the household's food expenditure decision is based on the equation

$$y = \beta_1 + \beta_2 x + e \tag{2.1}$$